

Automated Network Request Management – Project Documentation

Project Name	Automated Network Request Management
Plat form	ServiceNow
Project Type	IT Service Management (ITSM)
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PHASE 1: REQUIREMENT ANALYSIS & PLANNING

Project Overview

The **Network Request Management** project is designed to provide a structured and automated way for users to submit and manage network-related service requests through ServiceNow.

The main purpose of this project is to replace manual request handling with a centralized ServiceNow process. Users can submit their network requirements through a Service Catalog item, provide the required information, and track the request through its approval and fulfillment stages.

The project focuses on creating a simple request experience for end users while providing administrators and support teams with a controlled workflow for processing requests.

1.1 Business Objectives

Overview

The **Automated Network Request Management** project is designed to streamline and automate the complete lifecycle of network-related service requests using the **ServiceNow platform**.

The solution provides a **centralized and structured platform** where users can submit requests, provide required information, obtain approvals, receive notifications, and track requests until completion.

The main objective is to **reduce manual effort, improve processing efficiency, standardize request handling, and enhance the user experience**.

Key Business Objectives

Reduce Manual Effort

Automate repetitive request-processing activities and reduce dependency on emails, spreadsheets, and manual follow-ups.

Reduce Human Errors

Use mandatory fields, predefined options, validation rules, and automated workflows to improve data accuracy.

Accelerate Request Fulfilment

Automate routing, approvals, notifications, and task processing to reduce delays in request fulfilment.

Standardize Request Processing

Use standardized catalog items, variables, approval rules, workflows, and statuses for consistent request handling.

Automate Approval Processes

Route requests to the appropriate approvers and track approval decisions throughout the request lifecycle.

Provide Centralized Request Tracking

Maintain all network requests in ServiceNow and provide visibility into request numbers, statuses, approvals, and activities.

Improve End-User Experience

Provide a simple Service Portal interface for submitting, tracking, and managing network requests.

Improve Communication

Use automated email notifications to inform users, approvers, and support teams about important request updates.

Support IT and Security Compliance

Implement appropriate roles, permissions, approval processes, mandatory information, and standardized workflows.

Improve Auditability

Maintain activity history, approval records, system logs, and audit information for request tracking and verification.

Expected Business Benefits

The solution provides the following benefits:

- Reduced manual processing and administrative effort.
- Faster network request fulfilment.
- Reduced data-entry errors.
- Standardized request and approval processes.
- Automated notifications and workflow activities.
- Centralized request tracking and visibility.

- Improved communication between stakeholders.
- Better end-user experience.
- Improved access control and compliance.
- Better auditability and reporting.

Overall Objective

The overall objective is to create a **centralized, automated, and standardized network request management solution in ServiceNow** that manages the complete lifecycle from **request submission to approval, fulfilment, notification, and closure**.

The solution improves operational efficiency while providing better visibility, control, and user experience for both end users and IT support teams.

1.2 Functional Scope

The **Functional Scope** defines the major ServiceNow functionalities required to automate the complete lifecycle of network-related service requests.

The solution covers **request submission, form configuration, validation, approval, workflow automation, notifications, tracking, and closure**.

Catalog Item Creation

A **Network Request** catalog item is created in ServiceNow as the primary entry point for users.

It includes:

- Requester information.
- Network request details.
- Type of connection.
- Network configuration details.
- Business justification.
- Required mandatory fields.

Form Design and Customization

The request form is designed to capture the required information using appropriate variables and field types.

Dynamic form behavior is configured to **show, hide, or make fields mandatory** based on user selections. This improves data accuracy and user experience.

Data Validation

Mandatory fields, predefined choices, and validation rules ensure that users provide complete and accurate information before submitting requests.

Approval Routing

Requests requiring authorization are automatically routed to the appropriate approver.

The approval process supports:

- Pending approval.
- Approval.
- Rejection.

Approval status is tracked throughout the request lifecycle.

Flow Designer Automation

ServiceNow Flow Designer automates the request lifecycle.

The flow can:

1. Trigger when a request is submitted.
2. Validate request information.
3. Route the request for approval.
4. Process approval or rejection.
5. Send notifications.
6. Update request and task statuses.
7. Complete and close the request.

Email Notifications

Automated notifications keep users, approvers, and support teams informed about important events such as:

- Request submission.
- Approval or rejection.
- Assignment and status updates.
- Request completion and closure.

Request Tracking

Each request receives a unique request number that allows users and support teams to track its progress.

Important information includes **requester, request type, current status, approval status, assigned team, and completion details**.

Request Lifecycle Management

The complete lifecycle is managed through ServiceNow:

Submission → Validation → Approval → Processing → Fulfilment → Notification → Closure

Functional Scope Summary

The project includes:

- Network Request catalog item.
- Variables and form configuration.
- Dynamic fields and validation.
- Approval routing.
- Flow Designer automation.
- Email notifications.
- Request tracking and status management.
- Fulfilment and closure.
- Audit and activity tracking.

These functionalities provide a **structured and automated solution for managing network requests from submission to final closure**.

1.3 Approval Use Cases

Approval processing is an important part of the Network Request Management process. The project considers different approval scenarios depending on the nature of the request. **Manager Approval**

Standard network requests can be routed for manager approval before further processing. This provides an additional level of authorization before the request is fulfilled. **Network Security Approval**

Requests that involve sensitive network-related activities may require additional review from the appropriate network or security team.

This ensures that requests requiring additional attention can follow an appropriate approval process. **Group Approval**

Department-specific or team-specific requests can be routed to the appropriate group for review and approval.

This allows the request process to accommodate different organizational requirements.

1.4 Stakeholder Mapping

Stakeholder mapping identifies the key users and teams involved in the **Automated Network Request Management** solution.

The main stakeholders are **End Users, IT Administrators, Network Fulfilment Team, and Approvers**.

Stakeholder Details

Stakeholder	Role	Needs / Expectations	Impact of Automation
End Users (Requesters)	Submit network requests through the ServiceNow Service Portal.	Simple submission, fast processing, and status visibility.	Faster fulfilment, better tracking, and fewer manual follow-ups.

Stakeholder	Role	Needs / Expectations	Impact of Automation
IT Administrators	Manage ServiceNow configurations, workflows, catalog items, and access.	Reliable automation, easy maintenance, and controlled configuration.	Reduced administrative work and easier maintenance.
Network Fulfilment Team	Reviews approved requests and performs network fulfilment activities.	Complete request information, clear tasks, and standardized processes.	Automated task generation, fewer errors, and reduced repetitive work.
Approvers	Review and approve network requests according to defined policies.	Clear information, proper routing, and quick approval processing.	Automated approval routing, better tracking, and improved auditability.

Stakeholder Responsibilities

End Users

- Submit accurate network request information.
- Select the appropriate request type.
- Monitor request status.
- Provide additional information when required.

IT Administrators

- Manage catalog items, variables, and workflows.
- Configure Flow Designer, UI Policies, and notifications.
- Manage roles and access permissions.
- Monitor and troubleshoot the system.

Network Fulfilment Team

- Review approved requests.
- Perform required network activities.
- Update request and task statuses.
- Complete and close fulfilment tasks.

Approvers

- Review request details and business justification.
- Approve or reject requests.
- Provide comments when required.

- Ensure requests follow organizational policies.

Stakeholder Interaction

The main interaction flow is:

End User → ServiceNow Portal → Approval → Network Fulfilment → Completion → Notification

The automated workflow connects stakeholders and moves requests through the required stages with minimal manual coordination.

1.5 Execution Roadmap

The **Execution Roadmap** defines the major activities required to implement the **Automated Network Request Management** solution in ServiceNow.

The project is divided into five milestones:

Catalog Creation → Form Setup → Approval Integration → Testing → Deployment

Each milestone focuses on a specific stage of the implementation.

Milestone 1: Catalog Creation

Objective

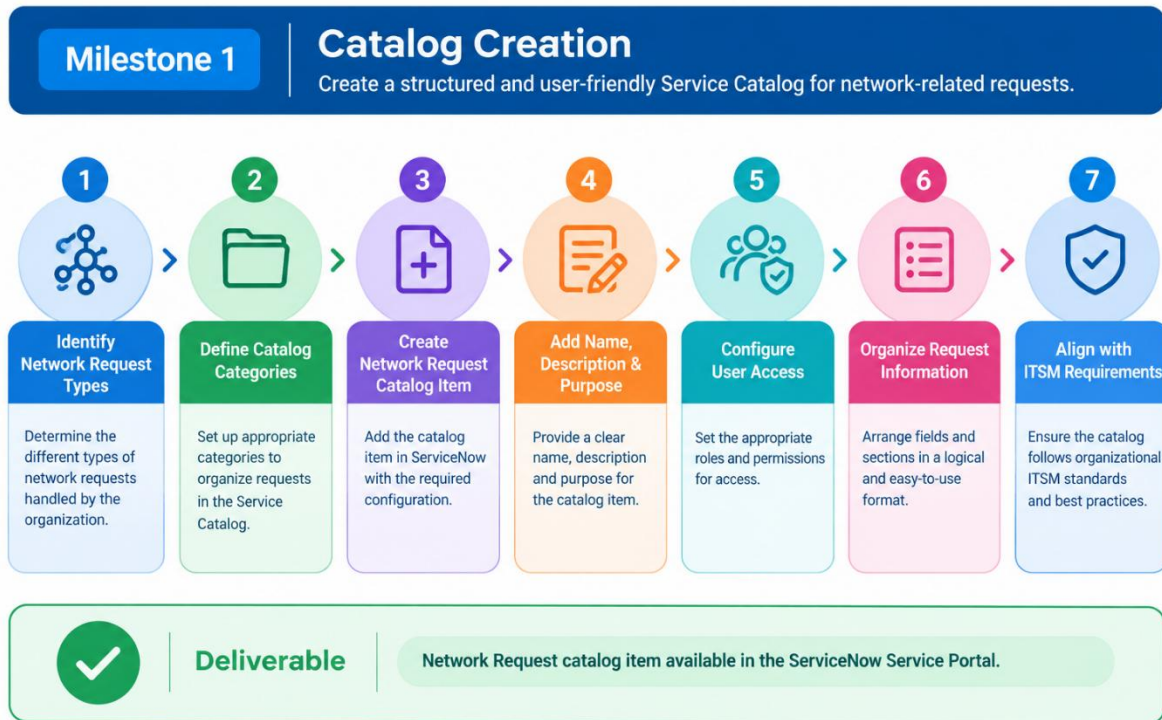
Create a structured and user-friendly Service Catalog for network-related requests.

Key Activities

- Identify network request types.
- Define catalog categories.
- Create the **Network Request** catalog item.
- Add a clear name, description, and purpose.
- Configure appropriate user access.
- Organize request information logically.
- Align the catalog with ITSM requirements.

Deliverable

Network Request catalog item available in the ServiceNow Service Portal.



Milestone 2: Form Setup

Objective

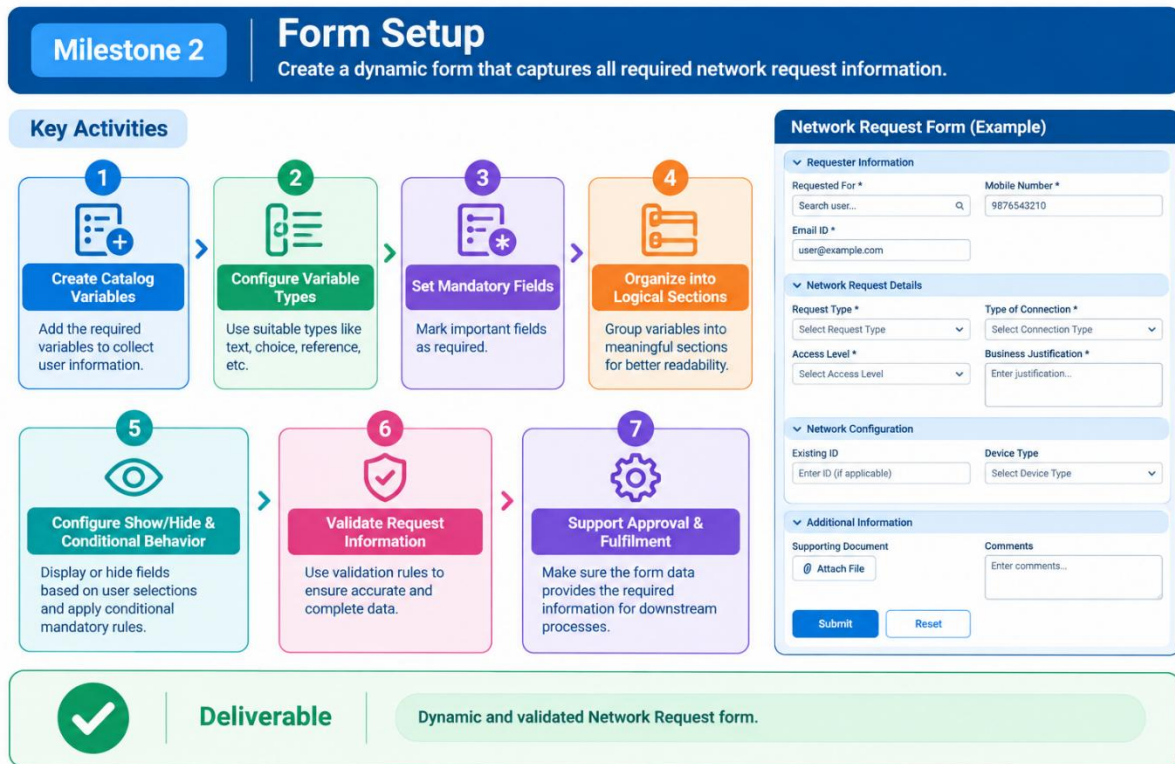
Create a dynamic form that captures all required network request information.

Key Activities

- Create required catalog variables.
- Configure appropriate variable types.
- Set mandatory fields.
- Organize variables into logical sections.
- Configure show/hide and conditional field behavior.
- Validate request information.
- Ensure form data supports approval and fulfilment.

Deliverable

Dynamic and validated Network Request form.



Milestone 3: Approval Integration

Objective

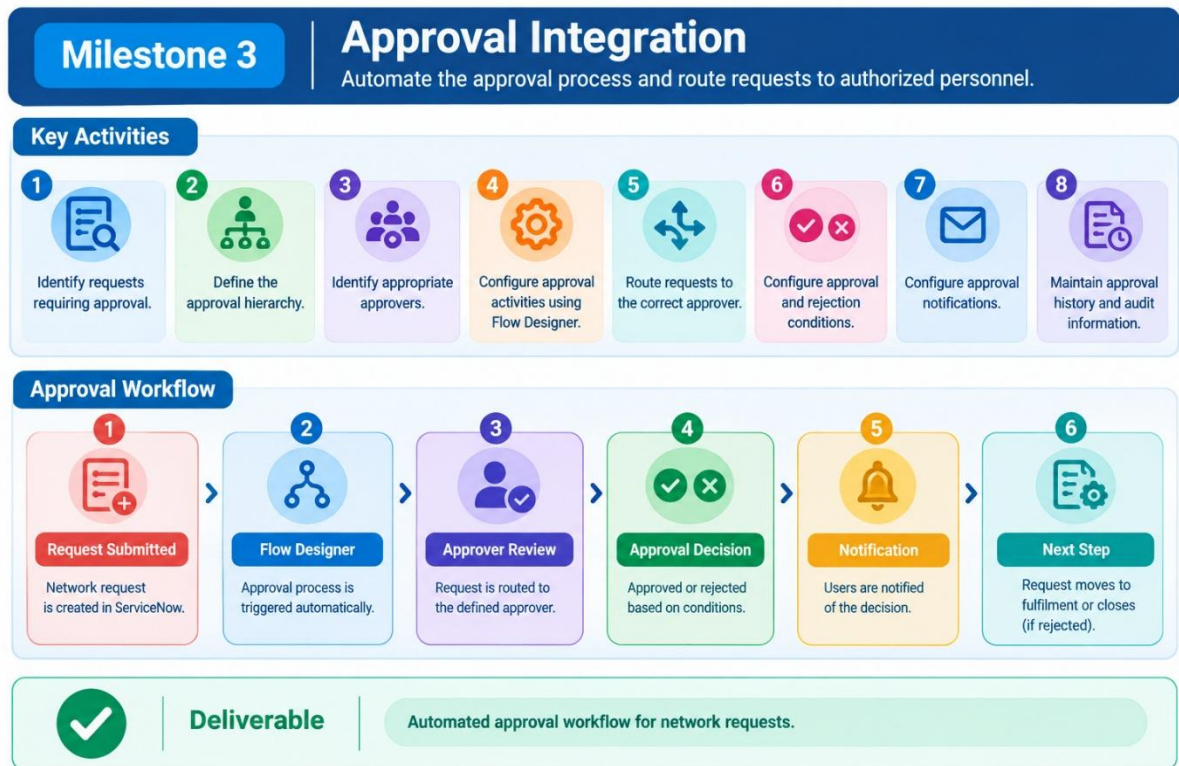
Automate the approval process and route requests to authorized personnel.

Key Activities

- Identify requests requiring approval.
- Define the approval hierarchy.
- Identify appropriate approvers.
- Configure approval activities using **Flow Designer**.
- Route requests to the correct approver.
- Configure approval and rejection conditions.
- Configure approval notifications.
- Maintain approval history and audit information.

Deliverable

Automated approval workflow for network requests.



1.4.5

Milestone 4: Testing



Objective

Validate the complete request lifecycle and ensure all configurations work correctly.



Key Activities

- Test catalog items, variables, mandatory fields, and UI Policies.
- Test Flow Designer and approval configurations.
- Perform end-to-end request testing.
- Verify request creation, task generation, approvals, notifications, and closure.
- Conduct User Acceptance Testing (UAT).
- Record and resolve identified defects.



Test Flow



Deliverable

Validated and tested end-to-end request lifecycle.

Test Flow

Request Submission → Validation → Approval → Fulfilment → Notification → Closure

Deliverable

Validated and tested end-to-end request lifecycle.

Milestone 5: Deployment

Objective

Deploy the tested solution and make it available to authorized users.

Key Activities

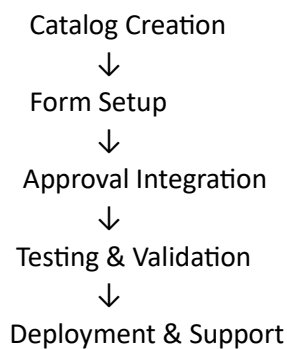
- Prepare the deployment checklist.
- Review and validate configurations.
- Migrate approved configurations.
- Verify roles and access.
- Perform post-deployment testing.
- Monitor the system after deployment.
- Provide basic user and support-team guidance.
- Establish post-deployment support.



Deliverable

Deployed and operational Network Request Management solution.

Execution Flow



Roadmap Summary

Milestone	Primary Focus	Key Deliverable
1. Catalog Creation	Service Catalog	Network Request catalog item
2. Form Setup	Request Form	Dynamic and validated form
3. Approval Integration	Automation	Automated approval workflow

Milestone	Primary Focus	Key Deliverable
4. Testing	Quality Validation	Tested request lifecycle
5. Deployment	Production Readiness	Operational solution

The execution roadmap provides a **structured approach from catalog creation to deployment**, ensuring that the solution is properly configured, tested, and ready for operational use.

1.6 Project Roadmap

The project follows a structured implementation approach.

First, the business requirements are identified and converted into functional requirements. The required ServiceNow catalog item and form fields are then planned based on these requirements.

After the form design is finalized, the approval and automation requirements are considered. The implementation is organized into separate milestones so that each part of the solution can be developed and tested independently.

Phase 1 Summary

Phase 1 – Requirement Analysis & Planning establishes the foundation for the Network Request Management solution.

The phase defines **what the system needs to accomplish, who will use it, what information must be collected, how approvals should work, and how the implementation will progress.**

PHASE 2: BACKEND DEVELOPMENT & CONFIGURATIONS

2.1 Data Architecture

Description

The **Automated Network Request Management** solution uses a structured ServiceNow data architecture to store and manage network-related request information.

The architecture combines **standard ServiceNow tables and custom tables/fields** to support request submission, approval, fulfilment, tracking, and closure.

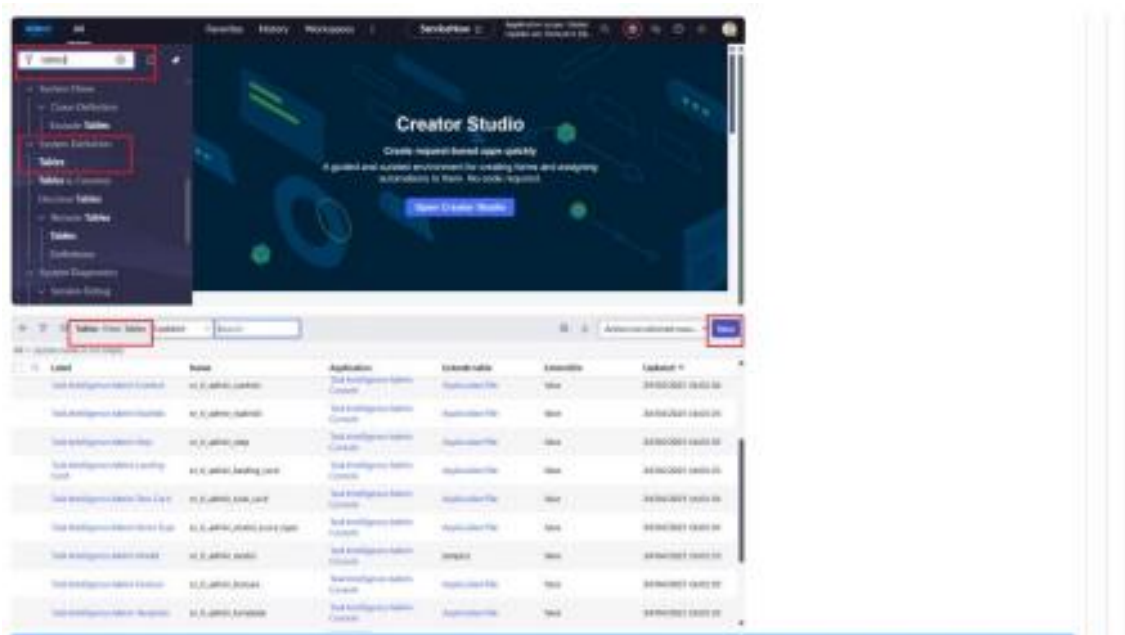
The data model can include:

- Request number
- Requested for
- Request type
- Type of connection
- Access level
- Business justification
- Approval status
- Request status

- Assignment information
- Related tasks
- Completion details

Custom fields are configured with appropriate field types, mandatory settings, reference fields, and choice values. UI Policies and related UI logic can be used to control field visibility and mandatory behavior.

The architecture also supports **list views and related lists** for monitoring requests, approvals, tasks, and statuses.



Activity 1: Creation of DatabaseTable

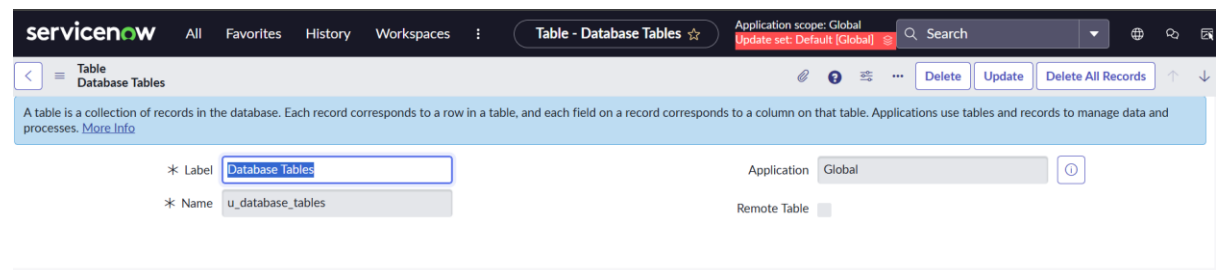
Objective: Create a custom table to store network-related information.

Field	Configuration
Label	Database Tables
Name	u_database_table
Purpose	Store network database/request information

Steps

1. Log in to ServiceNow.
2. Navigate to **All → System Definition → Tables**.
3. Click **New**.
4. Enter the Label: **Database Tables**.
5. Verify the table name: **u_database_table**.

6. Configure required settings.
7. Click **Submit/Save**.
8. Verify that the table was created successfully.



Activity 2: Creation of Fields

Objective: Create and organize fields required to capture network request information.

Steps

1. Navigate to **All → System Definition → Tables**.
2. Search for **Database Tables**.
3. Open the table.
4. Add the required fields/columns.
5. Select suitable field types such as **String, Choice, Reference, Integer, Date/Time, or Boolean**.
6. Mark required fields as **Mandatory** where applicable.
7. Open **Form Design** and arrange the fields in the required order.
8. Save the configuration.
9. Open a sample record and verify the fields.

Auto Number Configuration

1. Open the table configuration.
2. Locate the **Columns** section.
3. Configure the **Auto Number** field.
4. Enter the required **Prefix** and **Number format**.
5. Save the configuration.

Q	Column label	Type	Reference	Max length	Default value	Display
X	Mobile Number	String	(empty)	40		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
	Updated	Date/Time	(empty)	40		false
X	Total Amount	String	(empty)	40		false
	Created by	String	(empty)	40		false
X	Type of Connection	String	(empty)	40		false
X	Mode of Payment	String	(empty)	40		false
	Created	Date/Time	(empty)	40		false
X	Requested For	String	(empty)	40		false
X	Approval Status	String	(empty)	40		false
X	DataBase Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
X	Customer Address	String	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
X	Assignment to	Reference	User	32		false
	Insert a new row...					

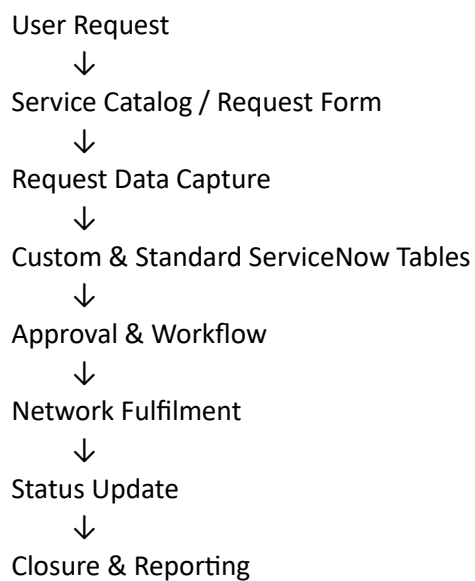
Example: NET0001, NET0002, NET0003

Verification

Open a **Network Database** record and verify that:

- Required fields are displayed.
- Fields are in the correct order.
- Sample data can be saved.
- Auto Number is generated correctly.

Data Architecture Flow



2.2 Data Architecture Benefits

- Centralized request information.
- Structured and consistent data.
- Reduced data-entry errors.
- Easier request tracking.
- Support for automated approvals.
- Improved visibility and reporting.
- Better auditability.
- Controlled access.
- Scalable data structure.

2.3 Approval Request

The **Approval Request** configuration connects the Network Database record with ServiceNow approval records.

This allows authorized users to view approval information directly from the Network Database form and monitor the approval lifecycle.

Activity 1: Creation of Relationship

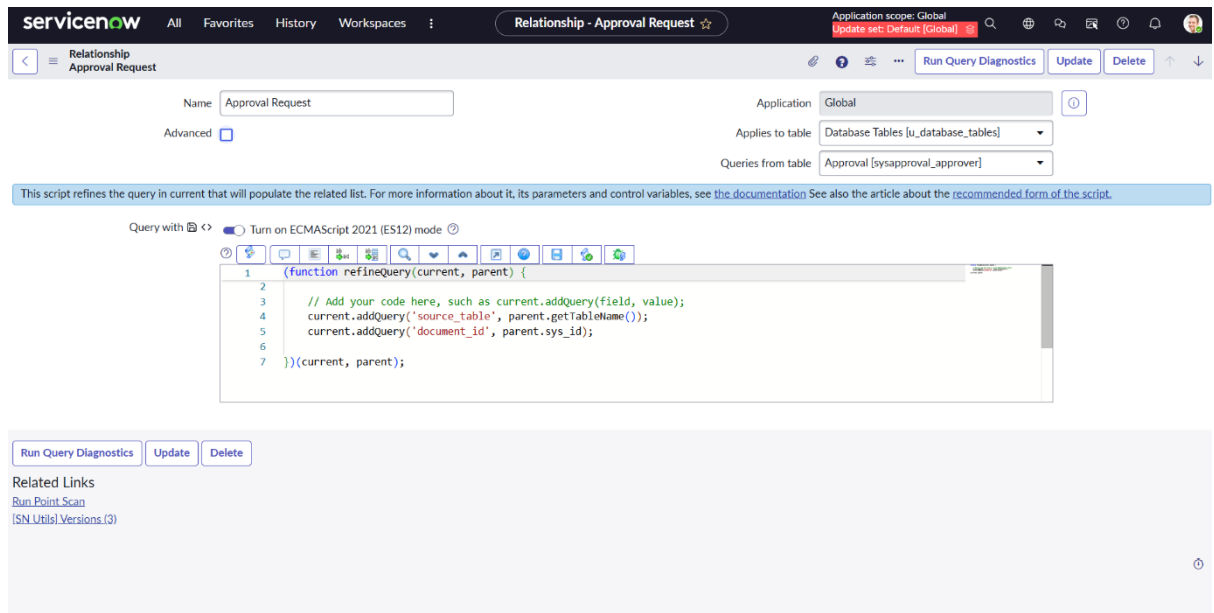
Objective: Create a relationship between the Database Table and the appropriate approval table.

Steps

1. Navigate to **All → System Definition → Relationships**.
2. Click **New**.
3. Configure the relationship:

Field	Value
Name	Approval Request
Applies to Table	Database Table
Queries from Table	Appropriate approval table, such as sysapproval_approver
Active	True

4. Review the configuration.
5. Click **Submit/Save**.

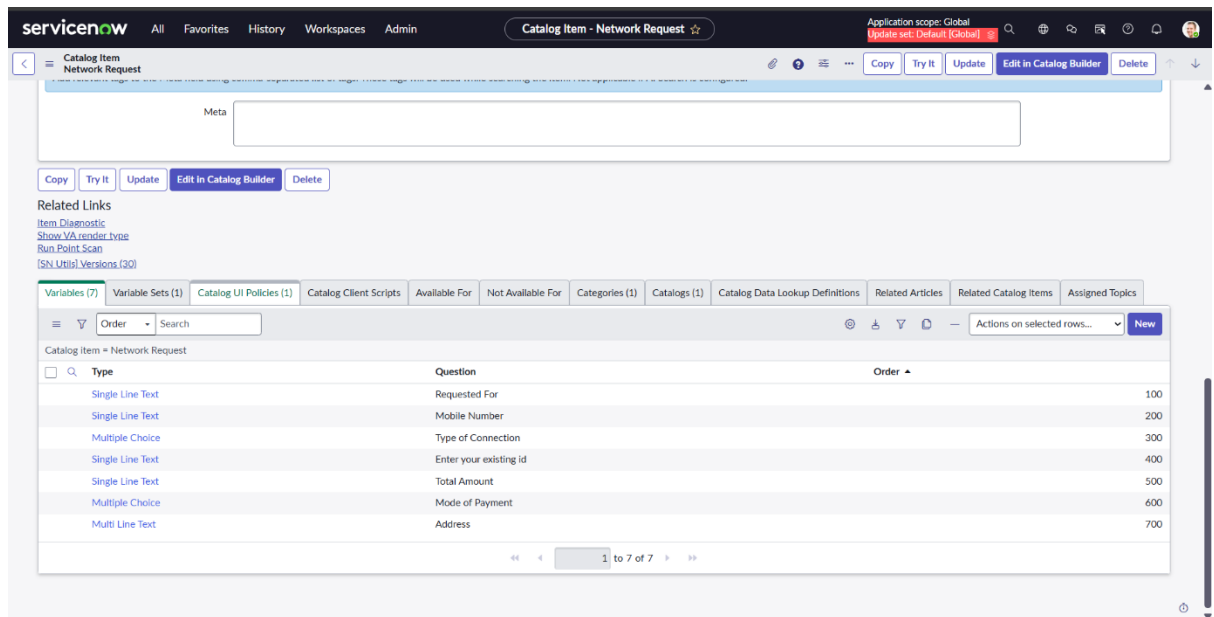


Activity 2: Add Related List

Objective: Display approval records on the Network Database form.

Steps

1. Open a **Network Database** record.
2. Open **Form Design** or the available form configuration option.
3. Locate **Related Lists**.
4. Add **Approval Request**.
5. Arrange it in the required position.
6. Click **Save**.
7. Reopen the record and verify the related list.



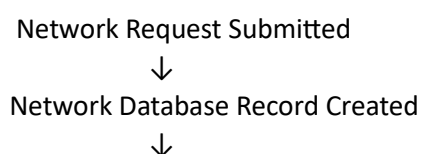
Activity 3: Configure Approval Records

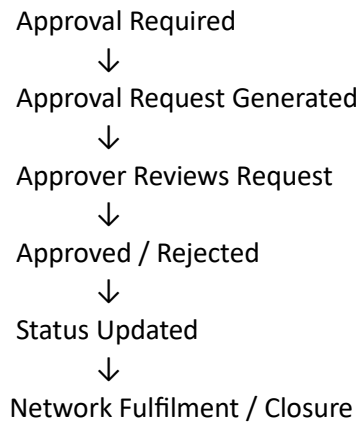
Objective: Verify that approval records are correctly associated with the network request.

Steps

1. Submit a Network Request requiring approval.
2. Allow the configured Flow/Workflow to create the approval.
3. Open the corresponding **Network Database** record.
4. Check the **Approval Request** related list.
5. Verify:
 - Approver
 - Approval state
 - Request reference
 - Date/time
 - Comments, if applicable
6. Approve or reject the request.
7. Verify that the approval status is updated.

Approval Request Flow





Overall Outcome

The backend configuration provides a **structured data foundation and integrated approval process** for managing network requests throughout their lifecycle.

Phase 3: UI/UX Development & Customization

3.1 Interface Design

The Interface Design phase focuses on creating a simple and user-friendly ServiceNow interface for submitting and managing network requests.

The solution includes **Service Catalog forms, variables, variable sets, UI Policies, and Service Portal components**. Dynamic behavior is used to show, hide, or make fields mandatory based on user selections.

Main Activities

1. Service Catalog Creation
2. Variables Configuration
3. Variables Creation
4. Variable Set Configuration

Activity 1: Creation of Service Catalog Item

Objective

Create a **Network Request** catalog item through which users can submit network-related requests.

Navigation

Application Navigator → Service Catalog → Maintain Items

Steps

1. Log in to ServiceNow.
2. Open **Maintain Items**.
3. Click **New**.

4. Configure:

- **Name:** Network Request
- **Catalog:** Service Catalog
- **Category:** Network
- **Short Description:** Network Request Management

5. Click **Save/Submit**.

6. Verify the catalog item.

Catalog Item - Network Request

Catalog Items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

NameNetwork RequestApplicationGlobalActiveFulfillment automation levelUnspecifiedState-- None --Checked out-- None --OwnerSystem Administrator

Item DetailsProcess EnginePicturePricingPortal SettingsShort descriptionNetwork RequestDescription<>

B I U ↶ ↷ Verdana 8pt
⋮ ⋮ ⋮ ⓘ

Press Alt+O for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Not applicable if AI Search is configured.

Meta

CopyTry ItUpdateEdit in Catalog BuilderDelete

Activity 2: Variables Configuration

Objective

Configure variables required to collect network request information.

Steps

1. Open the **Network Request** catalog item.
2. Open the **Variables** related list.
3. Click **New**.
4. Select the required variable type.
5. Configure **Question, Name, Order, Mandatory, Read-only, and Auto-population** where required.

- Save the variable.

Activity 3: Variables Creation

Variable	Type	Value / Purpose
Requested For	Reference/Text	User information
Mobile Number	Single Line Text	User mobile number
Type of Connection	Multiple Choice	New / Existing
Existing ID	Single Line Text	Existing connection ID
Total Amount	Single Line Text	Applicable amount
Mode of Payment	Multiple Choice	UPI / CARD
Address	Single Line Text	User address

Example: When **Type of Connection** = **Existing**, the Existing ID field can be displayed using a UI Policy.

Catalog Variables

Variables created in the Network Request catalog item.

S.No	Variable Name	Type	Choices / Values
1	Requested For	Single Line Text	—
2	Mobile Number	Single Line Text	—
3	Type of Connection	Multiple Choice	New / Existing
4	Enter your Existing ID	Single Line Text	—
5	Total Amount	Single Line Text	—
6	Mode of Payment	Multiple Choice	UPI / CARD
7	Address	Single Line Text	—

Network Request

Please fill in the details to submit your request.

- Requested For *
- Mobile Number *
- Type of Connection *
- Enter your Existing ID
- Total Amount *
- Mode of Payment *
- Address *

Submit

Activity 4: Variable Set Configuration

Objective

Create a reusable **Requester Information** variable set to organize user-related fields.

Steps

- Open the catalog item.

2. Locate **Variable Sets**.
3. Create a new variable set named **Requester Information**.
4. Add the required variables.
5. Configure their order and types.
6. Apply the variable set to **Network Request**.
7. Save and verify.

The screenshot shows the ServiceNow interface for configuring a Variable Set named 'Requester Information'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header indicates 'Variable Set - Requester Information' with an application scope of 'Global'. The configuration form includes fields for Title, Name, Order, Type, Display title, Layout, and Description. Below the form, a table lists the variables included in the set, their types, questions, and order.

Name	Type	Question	Order
opened_on_behalf_of	Reference	Opened on behalf of	100
emailid	Single Line Text	EmailID	200
user_name	Single Line Text	User Name	300
phone_number	Single Line Text	Phone Number	400

Requester Information Variables

Variable	Type	Configuration
Opened on behalf of	Reference	User [sys_user]
Email ID	Single Line Text	Auto-populate
User Name	Single Line Text	Auto-populate
Phone Number	Single Line Text	Auto-populate

Auto-Population Flow

Opened on behalf of → User selected → Email/User Name/Phone retrieved automatically

This reduces duplicate data entry and improves data accuracy.

Note: Follow the same procedure to auto populate values for all other variables

3.2 UI/UX Design Flow

Service Portal → Network Request → Requester Information → Network Request Variables → Dynamic Field Behavior → Validation → Submission → Approval/Fulfilment

UI/UX Benefits

- Simple and structured request submission.
- Reduced unnecessary data entry.
- Dynamic display of relevant fields.
- Improved data accuracy.

- Automatic requester information.
- Better Service Portal user experience.
- Easy integration with approval and fulfilment workflows.

3.3 Navigation Flow

Steps

1. Log in to the ServiceNow PDI.
2. Open the instance URL.
3. Add **/sp** to open the Service Portal.
4. Search for **Network Requests**.
5. Open the **Network Request** catalog item.
6. Enter the required details.
7. Complete mandatory fields.
8. Click **Submit**.
9. ServiceNow generates a unique request number.
10. Configured email notifications are sent to relevant users or teams.

The screenshot shows the 'Catalog Item' form for 'Network Request'. The form includes a header with navigation links (Copy, Try It, Update, Edit in Catalog Builder, Delete) and a blue banner with instructions: 'Build and modify items faster with the improved Catalog Builder. Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. • Enter a Name and Short description to display for the item. • Enter a Price, approvals, variables, and other information as needed.'

The form fields are as follows:

- Name:** Network Request
- Application:** Global
- Active:** ☒
- Fulfillment automation level:** Unspecified
- Catalogs:** Service Catalog
- Category:** Network Standard Changes
- State:** -- None --
- Checked out:** -- None --
- Owner:** System Administrator

At the bottom, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'.

Navigation Flow

PDI Login → /sp Service Portal → Search Network Requests → Open Catalog Item → Enter Details → Submit → Request Number → Approval/Fulfilment → Email Notification → Tracking & Closure

3.4 UI Policy

A **Catalog UI Policy** dynamically controls catalog form fields based on user selections. It can make fields **visible, hidden, mandatory, or read-only**.

Scenario 1: Existing Connection

Condition:

Type of Connection = Existing

Action:

Enter your Existing ID → Visible = True

Scenario 2: Other Device**Condition:**

Types of Devices = Others

Action:

Please Specify → Visible = True

UI Policy Configuration

1. Open **Network Request** catalog item.
2. Navigate to **Catalog UI Policies**.
3. Click **New**.
4. Select **Applies to = Catalog Item**.
5. Select **Catalog Item = Network Request**.
6. Enter the condition.
7. Save the UI Policy.
8. Open **UI Policy Actions**.
9. Select the required variable.
10. Set **Visible = True**.
11. Save the action.

The screenshot shows the ServiceNow Catalog UI Policy configuration page. The title bar indicates the policy is 'Catalog UI Policy - Show variable when connection type is Existing'. The 'Applies to' dropdown is set to 'A Catalog Item', and the 'Catalog Item' is 'Network Request'. The 'Application' is 'Global' and the policy is 'Active'. The 'Short description' is 'Show variable when connection type is Existing'. The 'When to Apply' tab is selected, showing the 'Catalog Conditions' section. The condition is 'type_of_connection is Existing'. The 'Applies on' section has checkboxes for 'Catalog Item view' (checked), 'Catalog Tasks' (checked), and 'Requested Items' (unchecked). The 'On load' checkbox is checked, and the 'Reverse if false' checkbox is also checked. The 'Update' and 'Delete' buttons are at the bottom left.

Testing

Test Case 1

- Select **Type of Connection = Existing**.
- Verify **Existing ID** becomes visible.
- Enter the Existing ID.
- Change the connection type and verify the field behavior.

Test Case 2

- Select **Types of Devices = Others**.
- Verify **Please Specify** becomes visible.
- Enter the required details.
- Change the device type and verify the field behavior.

Final UI Policy Logic

Type of Connection = Existing → Show Existing ID

Types of Devices = Others → Show Please Specify

This configuration provides a **dynamic, user-friendly, and controlled request form** while displaying additional fields only when required.

Phase 4: Data Migration, Testing & Security

Overview

Phase 4 focuses on validating the **Automated Network Request Management** solution and ensuring that data, automation, access control, and system functionality work correctly before deployment.

The phase covers four major areas:

Automation → Access Control → QA Testing → Data Integrity

4.1 Automation

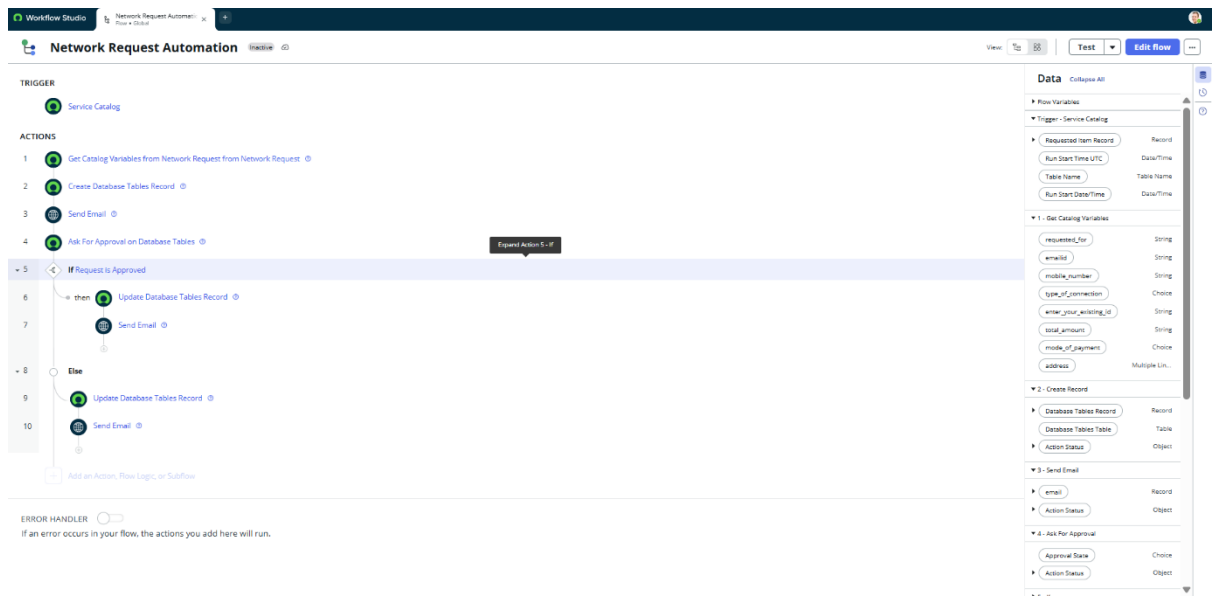
Objective

Verify that configured ServiceNow automation works correctly throughout the network request lifecycle.

Key Activities

- Test Flow Designer execution.
- Verify approval routing.
- Validate automated notifications.
- Check request and task status updates.
- Verify approved and rejected request paths.

- Confirm automated fulfilment activities.

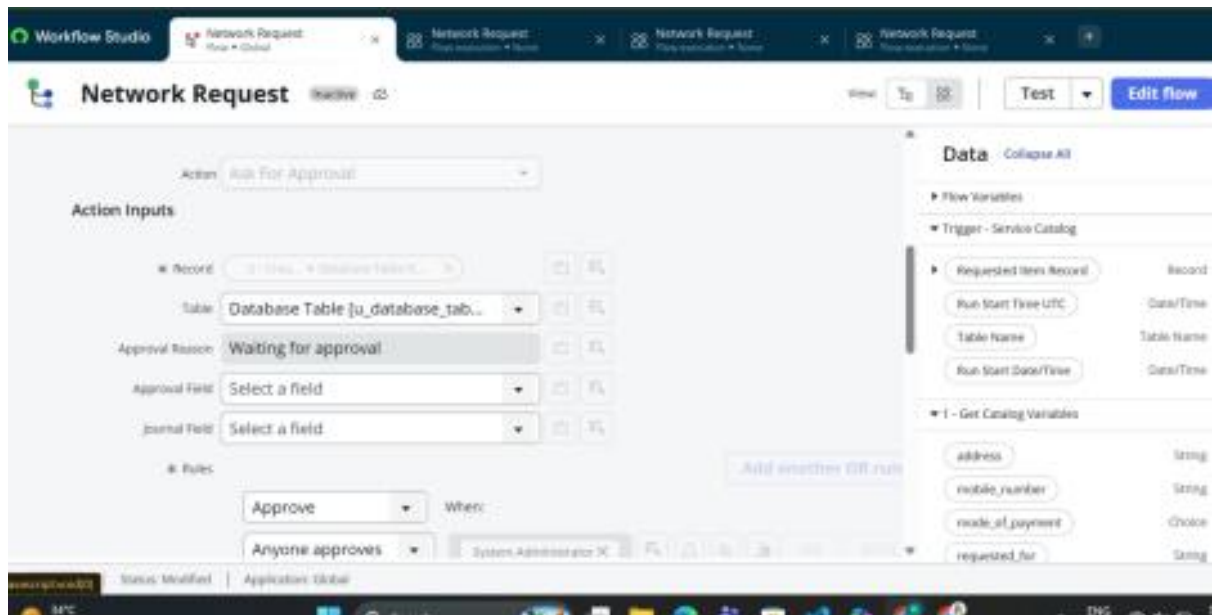


4.2 Access Control

Ensure that users can access only the information and functions required for their roles.

Key Activities

- Verify user roles and permissions.
- Test requester access.
- Validate administrator access.
- Check approver access.
- Verify network team access.
- Confirm restricted information is protected.



4.3 QA Testing

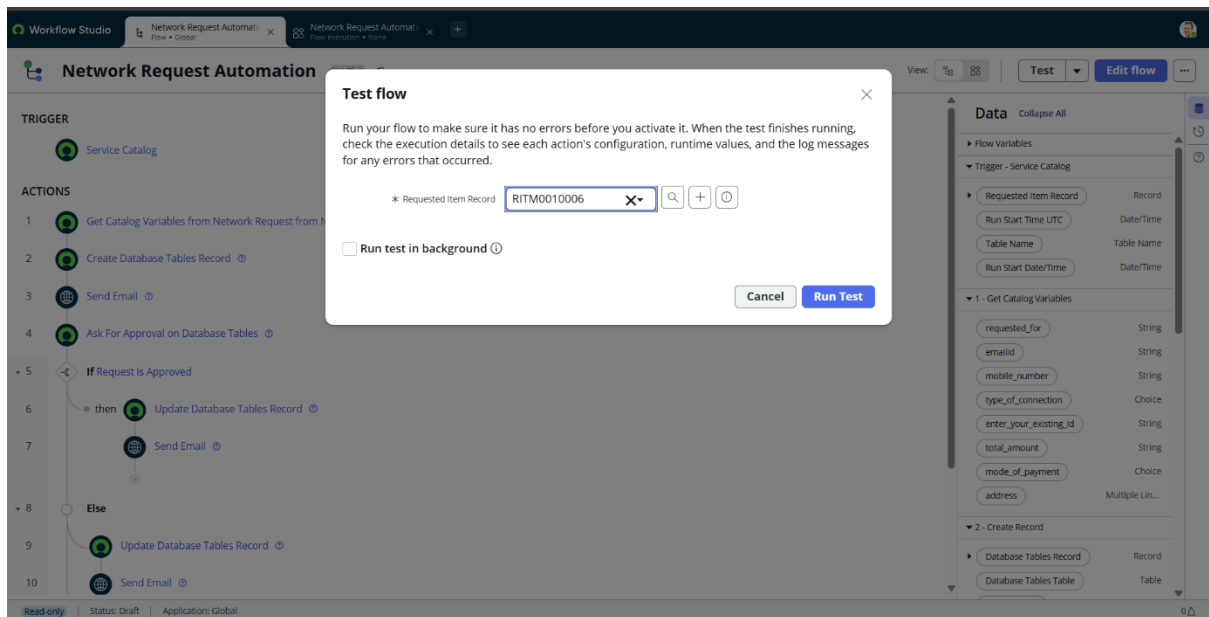
Validate the complete functionality of the solution before deployment.

Key Activities

- Test catalog item and variables.
- Verify mandatory fields and UI behavior.
- Test approval workflows.
- Validate notifications.
- Perform end-to-end testing.
- Conduct User Acceptance Testing (UAT).
- Record and resolve defects.

Test Flow

Request Submission → Validation → Approval → Fulfilment → Notification → Closure



4.4 Data Integrity

Ensure that network request data is accurate, complete, and consistently maintained.

Key Activities

- Verify required fields.
- Validate stored request information.
- Check data captured from catalog forms.
- Verify request and approval status.
- Confirm related records and tasks.
- Check activity and audit information.

EXECUTION DETAILS		Network Request		Test Run - Completed	Open flow	Open context record
Show Action Details			State	Start Time		
1	Get Catalog Variables from Network Request	Core Action	Completed	2026-09-25-00:03:58	16ms	
2	Create Record	Core Action	Completed	2026-09-25-00:03:58	4ms	
3	Send Email		Completed	2026-09-25-00:03:58	47 ms	
4	Ask For Approval	Core Action	Completed	2026-09-25-00:03:58	680ms	
5	If Request Approved	Flow Logic	Evaluated - True	2026-09-25-00:04:25	24ms	
6	Update Record	Core Action	Completed	2026-09-25-00:04:25	4ms	
7	Send Email		Completed	2026-09-25-00:04:25	20ms	
8	Else	Flow Logic	Not Run			
9	Update Record	Core Action	Not Run			
10	Send Email		Not Run			

4.5 Phase 4 Summary

Phase 4 ensures that the solution is **functionally tested, securely controlled, and data-accurate** before moving to the deployment phase.

Automation → Access Control → QA Testing → Data Integrity → Deployment Readiness

Phase 5: Deployment, Documentation & Final Presentation

5.1 Troubleshooting

Troubleshooting was performed to identify and resolve configuration or execution issues during development and testing.

The main areas checked were **Flow Designer execution, email notifications, variable population, form behavior, approvals, and record updates.**

Flow Execution Logs

Verify that all Flow Designer actions execute correctly.

Steps

1. Open **Flow Designer**.
2. Open the **Network Request** flow.
3. Open **Execution Details**.
4. Select the execution related to the test request.
5. Review each action and identify any failed or skipped step.
6. Check inputs, outputs, approval conditions, and update actions.
7. Correct the configuration and run another test.

Areas Verified:

- Service Catalog Trigger
- Get Catalog Variables
- Create Record
- Send Email
- Ask for Approval
- If/Else
- Update Record

Email Logs

Verify that automated email notifications are generated correctly.

Steps

1. Open the appropriate **Email Logs** section.
2. Search for the Network Request notification.
3. Verify recipients, subject, and message body.

4. Check the notification status.
5. If an email is missing, review the related Flow or notification configuration.

Variable Population Issues

Variable issues were checked when values were missing or incorrectly transferred.

Common checks:

- Variable belongs to the correct variable set.
- Correct variable name and order.
- Auto-population configuration.
- Catalog Client Script logic.
- Get Catalog Variables mapping.
- Correct data pills used in Flow Designer.

Form Behaviour Testing

Verify that fields behave correctly according to UI Policies.

Steps

1. Open the **Network Request** catalog item.
2. Test the form in the Service Portal.
3. Select different options.
4. Verify field visibility, mandatory, and read-only behavior.
5. Compare the result with the configured UI Policy.

Examples:

Type of Connection = Existing → Existing ID becomes visible

Types of Devices = Others → Please Specify becomes visible

Troubleshooting Summary

Area	Tool/Method	Purpose
Flow Execution	Flow Execution Details	Identify failures and conditions
Email	Email Logs	Verify notifications
Variables	Variable Sets, Client Scripts, Flow Mapping	Check missing/incorrect values
Form Behaviour	UI Policies, Service Portal	Verify dynamic fields
Approval	Approval Records, Flow Details	Verify approval processing
Record Updates	Record History, Flow Details	Verify correct updates

Troubleshooting Process

Issue Identified → Reproduce → Check Logs/Configuration → Find Root Cause → Correct → Retest → Verify → Document

5.2 Adherence to Timelines

The project followed a structured **5-phase implementation roadmap**, with each phase providing the foundation for the next.

Project Roadmap

Phase 1: Requirement Analysis & Planning

- Business objectives
- Functional requirements
- Stakeholder identification
- Scope definition
- Project roadmap

Phase 2: Backend Development & Configurations

- Custom tables
- Fields and auto-numbering
- Relationships
- Approval-related configurations

Phase 3: UI/UX Development & Customization

- Network Request catalog item
- Variables and variable sets
- UI Policies
- Conditional field behavior
- Auto-population

Phase 4: Data Migration, Testing & Security

- Flow Designer automation
- Record creation
- Approvals
- Email notifications
- ACLs
- QA testing

- Data integrity validation

Phase 5: Deployment, Documentation & Final Presentation

- Troubleshooting
- Flow and email verification
- Form testing
- Documentation
- Screenshots and diagrams
- Final presentation

Timeline Summary

Phase	Milestone	Deliverables
Phase 1	Requirement Analysis	Requirements, scope, roadmap
Phase 2	Backend Development	Tables, fields, relationships
Phase 3	UI/UX Development	Catalog, variables, UI Policies
Phase 4	Testing & Security	Automation, approvals, ACLs, QA
Phase 5	Deployment & Documentation	Troubleshooting, documentation, presentation

Execution Sequence

Requirement Analysis → Backend Configuration → UI/UX Development → Automation & Security → Testing → Troubleshooting → Documentation → Final Presentation

5.3 Innovation

The **Automated Network Request Management** solution uses a configuration-driven approach to automate the network request lifecycle in ServiceNow.

The implementation mainly uses standard ServiceNow features, reducing unnecessary custom development and making the solution easier to maintain.

Key Innovative Features

1. End-to-End Automation

Flow Designer automates:

- Request submission
- Catalog variable retrieval
- Network Database record creation
- Approval routing

- Email notifications
- Approval decision processing
- Record updates
- Fulfilment and closure

Automation Flow:

Request Submission → Get Catalog Variables → Create Record → Send Notification → Request Approval → Approval Decision → Update Record → Fulfilment / Closure

2. Configuration-Driven Development

The solution primarily uses **Flow Designer, UI Policies, Catalog Variables, and standard ServiceNow configurations** instead of extensive custom scripting.

Benefits include:

- Easier configuration
- Simplified troubleshooting
- Reduced maintenance
- Better knowledge transfer

3. Dynamic Approval Processing

Requests are routed to the appropriate approver based on configured conditions.

Request → Approval → Decision → Conditional Processing

This ensures required approvals are completed before fulfilment.

4. Dynamic Record Processing

Flow Designer automatically creates and updates Network Database records using data pills and conditions.

Catalog Request → Network Database → Approval → Fulfilment

5. Minimal Client-Side Customization

UI Policies and Catalog Client Scripts are used where required for:

- Auto-populating requester information
- Showing or hiding fields
- Making fields mandatory
- Controlling form behavior

6. Maintainability and Handoff

The solution supports easy maintenance through:

- Standard ServiceNow configurations

- Structured Flow Designer logic
- Limited scripting
- Documented variables and approval conditions
- Consistent naming
- Testing and troubleshooting documentation

Innovation Summary

Innovation Area	Implementation
Automation	End-to-end processing using Flow Designer
Approval	Dynamic approval routing
Record Creation	Automatic Network Database record creation
Notifications	Automated email notifications
UI Customization	UI Policies and minimal Client Scripts
Development	Configuration-driven approach
Maintainability	Standard and documented configurations

5.4 Functional Overview

The **Automated Network Request Management** solution provides a structured process for submitting, approving, processing, and tracking network requests through ServiceNow.

The catalog form captures required information, while Flow Designer manages approvals, notifications, record updates, and fulfilment.

Request Information

The Network Request catalog form captures the information required to process a request.

Variable	Purpose
Request Type	Identifies the requested network service
Business Justification	Captures the reason for the request
Requested For	Identifies the user for whom the request is submitted
Type of Connection	Identifies new or existing connection
Existing ID	Captures the existing connection ID when applicable

Variable	Purpose
Mode of Payment	Captures the selected payment method
Address	Captures required location information
Urgency	Indicates the processing priority

Approval Use Cases

1. Manager Approval

Standard requests can be routed to the appropriate manager.

Request Submitted → Validation → Manager Approval → Approved → Fulfilment

2. Network Security Approval

Sensitive network requests can be routed to appropriate security personnel.

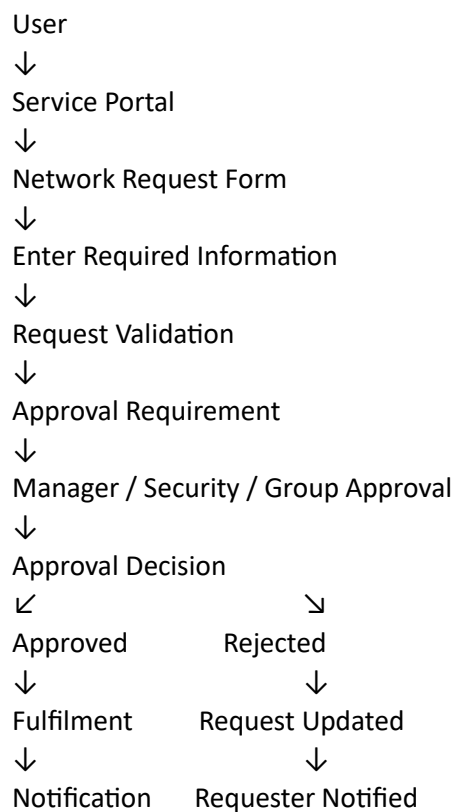
Sensitive Request → Security Review → Approval → Controlled Fulfilment

3. Group Approval

Department-specific requests can be routed to the relevant approval group.

Request → Identify Group → Group Approval → Approved → Fulfilment

Functional Workflow



↓

Closure

Functional Benefits

The solution provides:

- Structured request information
- Consistent data capture
- Automated approval routing
- Reduced manual intervention
- Improved request visibility
- Controlled processing of sensitive requests
- Clear audit history
- Standardized fulfilment

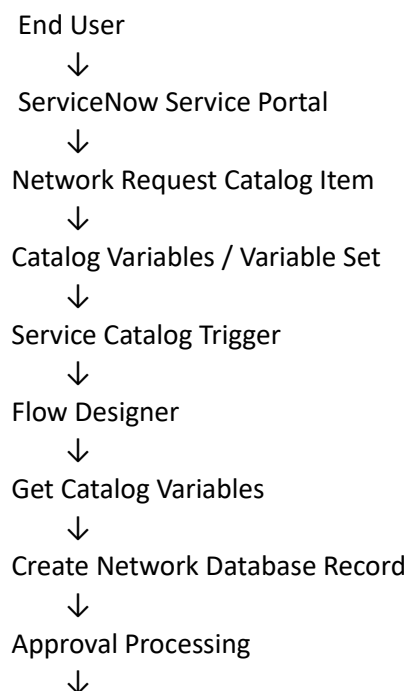
5.5 Technical Blueprint

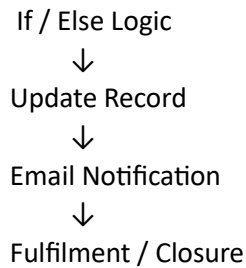
The **Technical Blueprint** describes the architecture and ServiceNow components used to implement the **Automated Network Request Management** solution.

It defines the relationship between the **Service Portal, Catalog Item, Variables, Custom Tables, Flow Designer, Approvals, Notifications, and Fulfilment**.

Technical Architecture

The overall architecture is:





Flow Designer Workflow

Flow Designer is the main automation component.

Main Components:

1. Service Catalog Trigger
2. Get Catalog Variables
3. Create Record
4. Send Email
5. Ask for Approval
6. If Condition
7. Update Record
8. Else Condition
9. Update Record

Flow:

Request Submitted → Get Variables → Create Record → Send Email → Ask for Approval → Approval Decision → Update Record → Fulfilment / Rejection

Screenshots of the Flow Designer should be included to show the trigger, actions, approval logic, conditions, and complete flow.

Variable-to-Field Mapping

Catalog variables are retrieved through **Get Catalog Variables** and mapped to fields in the Network Database table.

Mapping Flow:

Catalog Variable → Get Catalog Variables → Data Pill → Network Database Field

Catalog Variable	Target Field	Purpose
Requested For	Requested For	Identifies requester
Mobile Number	Mobile Number	Stores contact number
Type of Connection	Connection Type	Identifies connection type

Catalog Variable	Target Field	Purpose
Existing ID	Existing ID	Stores existing connection ID
Total Amount	Total Amount	Stores applicable amount
Mode of Payment	Payment Mode	Stores payment method
Address	Address	Stores location
Business Justification	Justification	Stores request reason
Urgency	Urgency	Indicates priority

Field names should be updated to match the exact fields configured in the ServiceNow PDI.

Custom Table Schema

The solution uses custom tables to store request and fulfilment information.

Network Database Table

Table Name: u_network_database

Purpose: Stores structured network request information.

Typical fields include:

- Request Number
- Requested For
- Request Type
- Connection Type
- Existing ID
- Business Justification
- Urgency
- Address
- Approval Status
- Request Status
- Created/Updated Date

Structure:

Network Request → Network Database Record → Request + Approval + Status Information

Network Task Table

Table Name: u_network_task

Purpose: Stores fulfilment-related task information.

Typical fields include:

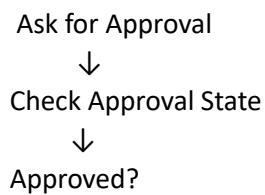
- Task Number
- Parent Request
- Assigned Group/User
- Task Description
- Status
- Priority
- Work Notes

Relationship:

Network Request → Network Database → Network Task → Fulfilment

Approval Condition Logic

The flow checks the approval result before continuing to fulfilment.



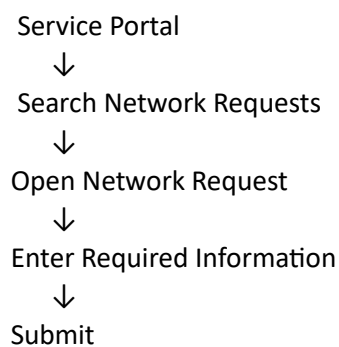
YES → Update Record → Continue to Fulfilment

NO → Update Record → Rejected → Notify Requester

This ensures that required approvals are completed before restricted processing.

Portal Configuration

The ServiceNow Service Portal provides the user interface.



The catalog form uses variables, mandatory fields, UI Policies, and conditional behavior.

Technical Component Summary

Component	Technical Purpose
Service Portal	User interface for submitting requests
Catalog Item	Defines the Network Request service
Catalog Variables	Capture request information
Variable Set	Organizes related variables
UI Policies	Control field behavior
Flow Designer	Automates the request lifecycle
Network Database	Stores request information
Network Task	Stores fulfilment tasks
Approval Action	Processes approvals
Email Action	Sends notifications
ACLs	Controls access
Audit History	Provides request traceability

Technical Blueprint Summary

The solution integrates the **Service Portal**, **Service Catalog**, **custom tables**, **Flow Designer**, **approvals**, **notifications**, and **security controls**.

Overall Architecture:

User Interface → Request Capture → Data Mapping → Record Creation → Approval → Conditional Processing → Fulfilment → Notification → Closure

This architecture provides a structured and maintainable foundation for future enhancements such as additional request types, approval rules, notifications, reports, and fulfilment tasks.

5.6 Setup Manual

The **Setup Manual** provides the steps required to recreate the **Automated Network Request Management** solution in a ServiceNow PDI. It covers tables, catalog configuration, variables, UI Policies, Flow Designer, approvals, ACLs, testing, and validation.

Prerequisites

Before starting, ensure:

- ServiceNow PDI is available.

- Administrator/developer access is available.
- Service Portal and Flow Designer are accessible.
- Test users are available for requester and approver testing.
- Required project fields and configurations are defined.

Implementation Sequence

Custom Tables → Fields → Relationships → Catalog Item → Variables → Variable Set → UI Policies → Flow Designer → Approvals → Notifications → ACLs → Testing → Validation

Custom Table Setup

Navigation:

All → System Definition → Tables

Network Database

Configuration	Value
Label	Network Database
Name	u_network_database
Purpose	Store network request information

Network Task

Configuration	Value
Label	Network Task
Name	u_network_task
Purpose	Store fulfilment task information

Create the required fields and verify the tables after saving.

Catalog Item Creation

Navigation:

All → Service Catalog → Catalog Definitions → Maintain Items

Steps

1. Click **New**.
2. Enter **Network Request**.
3. Select the required catalog and category.

4. Add the short description.
5. Configure required settings.
6. Save the catalog item.

Variable Configuration

Create variables required to capture request information.

Variable	Type	Example
Requested For	Reference/Text	User
Mobile Number	Single Line Text	Mobile number
Type of Connection	Choice	New / Existing
Existing ID	Single Line Text	Connection ID
Total Amount	Single Line Text	Amount
Mode of Payment	Choice	UPI / Card
Address	Single Line Text	Address
Business Justification	Multi Line Text	Request reason
Urgency	Choice	Low / Medium / High

Configure the **label, name, type, order, mandatory status, and choices** where required.

Variable Set Configuration

Create the **Requester Information** variable set where applicable.

Example variables:

- Opened on behalf of
- Email ID
- User Name
- Phone Number
- Location

Configure auto-population where required and verify that the variables appear correctly on the catalog item.

UI Policy Configuration

Configure Catalog UI Policies for conditional fields.

Example 1:

Type of Connection = Existing → Show Existing ID

Example 2:

Types of Devices = Others → Show Please Specify

Test the catalog form to verify the conditional behavior.

Flow Designer Setup

Navigation:

All → Flow Designer

Steps

1. Click **New → Flow**.
2. Enter **Network Request** as the flow name.
3. Add a suitable description.
4. Click **Build Flow**.
5. Configure the **Service Catalog Trigger**.
6. Add the required actions.

Flow Sequence**Approval Configuration**

Configure approval processing according to project requirements.

Possible approval types include:

- Manager approval
- Network security approval
- Department/group approval

Approved → Continue Processing

Rejected → Update Request / Notify Requester

ACL and Role Configuration

Navigation:

All → System Security → Access Control (ACL)

Review access for:

- Create
- Read
- Write
- Delete

Use appropriate roles and retain default ACLs if they meet the project requirements.

Test Submission Data

Use non-sensitive test data in the PDI.

Field	Test Value
Requested For	Test User
Mobile Number	Test Number
Type of Connection	Existing
Existing ID	TEST001
Total Amount	5000
Mode of Payment	UPI
Address	Test Address
Urgency	Medium
Business Justification	Network service required for testing

Test Execution

1. Open the ServiceNow Service Portal.
2. Go to **/sp**.
3. Search for **Network Requests**.
4. Open the Network Request catalog item.
5. Enter test information and submit.

6. Verify mandatory and conditional fields.
7. Record the generated request number.
8. Verify the Network Database record.
9. Verify Flow Designer execution.
10. Verify approval creation and processing.
11. Verify email notifications.
12. Verify status updates and final closure.

Validation Checklist

Area	Verification
Catalog Item	Available in Service Portal
Variables	Display correctly
Mandatory Fields	Prevent incomplete submission
UI Policies	Conditional fields work
Data Mapping	Values reach target record
Flow	Executes after submission
Record Creation	Network Database record created
Approval	Correct approver receives request
Email	Notifications generated
ACL	Access is controlled
Status	Request status updates correctly
Closure	Request reaches final state

Screenshot Documentation

Include screenshots of the major configurations:

1. Custom tables and fields
2. Catalog item
3. Variables and variable set
4. UI Policy
5. Flow Designer

6. Approval and email actions
7. ACL configuration
8. Service Portal form
9. Submitted request
10. Approval record
11. Final request status

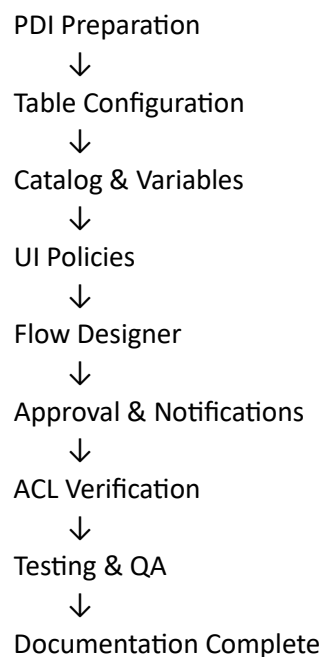
Each screenshot should have a short description.

Setup Completion Criteria

The setup is complete when:

- Tables and fields are configured.
- Catalog item and variables work correctly.
- UI Policies function correctly.
- Flow automation executes successfully.
- Approval and notifications work.
- ACLs are verified.
- Test submission is successful.
- Complete request lifecycle is validated.
- Required screenshots are documented.

Final Setup Flow



5.7 Project Demo Video Planning

Description

The **Project Demo Video** demonstrates the complete **Automated Network Request Management** lifecycle in ServiceNow, from request submission to approval, notification, fulfilment, and closure.

The demo covers three main roles:

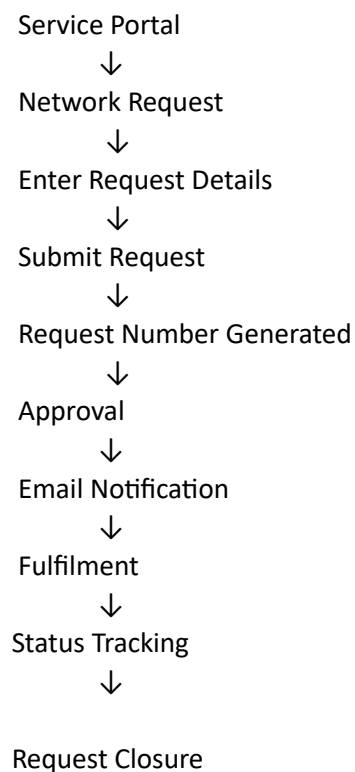
- **Requester**
- **Approver**
- **Network Fulfiller / IT Team**

Demo Objectives

The demo aims to:

- Demonstrate the Service Portal and Network Request catalog item.
- Show request submission and dynamic form behavior.
- Demonstrate automated approval processing.
- Verify email notifications.
- Show request status tracking.
- Demonstrate fulfilment and request closure.

Demo Flow



Scene 1 – Introduction

Screen: Project title – **Automated Network Request Management using ServiceNow**

Narration:

"This demonstration shows the automated network request lifecycle in ServiceNow, including submission, approval, notification, fulfilment, and closure."

Scene 2 – Service Portal

Role: Requester

Navigate to:

Service Portal → Network Services → Network Request

Demonstrate the catalog item and request form.

Scene 3 – Dynamic Request Form

Show important fields such as:

- Requested For
- Mobile Number
- Type of Connection
- Existing ID
- Total Amount
- Mode of Payment
- Address
- Business Justification
- Urgency

Example:

Type of Connection = Existing



Existing ID becomes visible.

This demonstrates the configured **UI Policy** and **dynamic form behavior**.

Scene 4 – Request Submission

Role: Requester

1. Enter test information.
2. Complete mandatory fields.
3. Click **Submit**.
4. Capture the generated request number.

Narration:

"After submission, ServiceNow creates the request and generates a unique request number for tracking."

Scene 5 – Record Creation

Role: IT Administrator / Fulfiller

Open the corresponding **Network Database** record and show:

- Request information.
- Request status.
- Approval information.
- Created record.

Narration:

"Flow Designer retrieves the catalog variables and automatically creates the Network Database record."

Scene 6 – Approval

Role: Approver

Show the approval request and demonstrate:

Request Submitted → Approval Generated → Approver Review → Approve/Reject

Narration:

"The request is automatically routed to the appropriate approver, who reviews the request and provides an approval decision."

Scene 7 – Email Notification

Show the generated notification with:

- Recipient
- Subject
- Request number
- Request status

Narration:

"Automated email notifications keep users informed about important request updates."

Scene 8 – Status Tracking and Fulfilment

Show the request lifecycle:

Submitted → Pending Approval → Approved → In Progress → Completed

After approval, demonstrate the fulfilment activity and update the request status.

Scene 9 – Request Closure

Role: Network Fulfiller / IT Administrator

Show:

- Fulfilment completion.
- Final status.
- Activity history.

Narration:

"After fulfilment is completed, the request is updated to the final closed or completed state."

Role-Based Demonstration

Role	Activities
Requester	Submit request, receive notifications, track status
Approver	Review and approve/reject requests
Network Fulfiller	Process approved requests and update status
IT Administrator	Verify records, flow execution, and notifications

Demo Video Timeline

Scene	Demonstration	Role
1	Project Introduction	Presenter
2	Service Portal Navigation	Requester
3	Request Form	Requester
4	Dynamic Field Behavior	Requester
5	Request Submission	Requester
6	Network Database Record	IT Admin
7	Approval	Approver
8	Email Notification	Requester / Approver
9	Status Tracking	Requester
10	Fulfilment	Network Fulfiller
11	Request Closure	Fulfiller / Admin

Scene	Demonstration	Role
12	Final Summary	Presenter

Final Demo Summary

The demo concludes with the complete lifecycle:

Submit Request → Create Record → Approval → Notification → Fulfilment → Status Tracking → Closure

Final Narration:

"This demonstration shows how ServiceNow automates the network request lifecycle using Service Portal, Catalog Items, Flow Designer, approvals, notifications, and status tracking."

5.8 Visual Demonstration

The **Visual Demonstration** section provides screenshot-based evidence of the major functionalities implemented in the **Automated Network Request Management** solution.

Screenshots are captured at important stages such as form behavior, approval processing, email notifications, Flow Designer execution, and request status changes. Short annotations and captions are used to explain each screenshot.

Form UI Behavior

Demonstrate the dynamic behavior of the Network Request catalog form.

Screenshots to Capture:

1. **Network Request Form** – Show the catalog item, request fields, and mandatory fields.
2. **Conditional Field** – Show **Type of Connection = Existing** and the **Existing ID** field becoming visible.
3. **Other Device Selection** – Show **Types of Devices = Others** and the **Please Specify** field becoming visible.

Approval Records

Show that submitted requests are correctly routed for approval.

Capture:

- Request reference.
- Approver.
- Approval state.
- Related request details.

Flow:

Request Submitted → Approval Created → Approver Review → Approved/Rejected

Email Notifications

Objective:

Demonstrate that notifications are generated at the required stages.

Capture:

- Recipient.
- Subject.
- Request number.
- Status.
- Message body.

Example:

Request Submitted → Submission Notification

Approval Completed → Approval Notification

Request Completed → Completion Notification

Demonstrate successful execution of the automated flow.

Open **Flow Designer** → **Execution Details** for the test request.

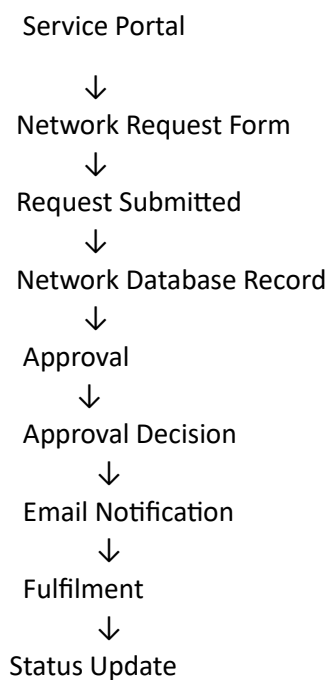
Execution Path:

Trigger → Get Catalog Variables → Create Record → Send Email → Ask for Approval → If Condition → Update Record

Highlight the successful actions, approval decision, condition path, and record update.

Request Lifecycle Visual

The complete lifecycle can be shown as





Request Closure

Screenshots should be provided for the major stages wherever practical.

Visual Annotation Standards

Use consistent annotations throughout the document:

- **Box/Outline:** Important field or configuration.
- **Arrow:** Shows process direction.
- **Numbered Label:** Identifies screen elements.
- **Caption:** Briefly explains the screenshot.
- **Highlight:** Shows the expected result.

Example Captions:

- **Figure 5.8.1:** Network Request catalog form.
- **Figure 5.8.2:** Conditional field behavior.
- **Figure 5.8.3:** Approval record for the Network Request.
- **Figure 5.8.4:** Flow Designer execution details.

Visual Evidence Checklist

Area	Evidence	Purpose
Form UI	Catalog form and dynamic fields	Verify form behavior
Variables	Variable values and mandatory fields	Verify data capture
Approval	Approval record and state	Verify approval routing
Email	Email notification/log	Verify communication
Flow	Execution Details	Verify automation
Database	Network Database record	Verify data mapping
Status	Request status/history	Verify lifecycle
Closure	Final request state	Verify completion

Recommended Screenshot Sequence

1. Service Portal Home
2. Network Request Catalog Item
3. Request Form

4. Dynamic Field Behavior
5. Submitted Request / Request Number
6. Network Database Record
7. Approval Record
8. Approved State
9. Email Notification
10. Flow Designer Execution Details
11. Request Status
12. Final Closed Request

This sequence provides visual evidence of the complete request lifecycle.

Expected Outcome

The visual demonstration should provide evidence of:

- Dynamic form behavior.
- Correct data capture.
- Automatic record creation.
- Approval processing.
- Email notifications.
- Flow execution.
- Status progression.
- Final request closure.

The annotated screenshots make the implementation easier to understand and support the project's **testing, documentation, and final presentation**.

5.9 Explanation Clarity

The **Automated Network Request Management** documentation follows a clear **What → Why → How → Outcome** structure. This helps technical and non-technical users understand the implementation and reproduce it in a ServiceNow PDI.

What – What Was Implemented?

The project documents the major ServiceNow components, including:

- Network Request Catalog Item
- Catalog Variables and Variable Sets
- Network Database Table

- UI Policies
- Flow Designer
- Approval Processing
- Email Notifications
- ACLs and Request Tracking

Why – Why Is It Required?

Each configuration is explained based on its purpose.

- **Catalog Item:** Provides a standard interface for submitting network requests.
- **UI Policies:** Display relevant fields based on user selections.
- **Flow Designer:** Automates record creation, approvals, notifications, and updates.

How – How Was It Implemented?

Each activity includes:

1. Navigation path
2. Configuration values
3. Implementation steps
4. Screenshot
5. Verification

For example:

All → Service Catalog → Maintain Items → New

Configure the Network Request catalog item with the required catalog, category, and description, then save and verify it.

Outcome – What Was Achieved?

The implementation provides:

- Standardized request submission
- Conditional form behavior
- Automatic record creation
- Approval processing
- Email notifications
- Request status tracking
- Request closure

Documentation Structure

Each major activity follows:

Objective → What → Why → Navigation → Configuration → Screenshot → Verification → Outcome

Screenshot Clarity

Screenshots should include:

- Figure number and caption
- Highlighted fields or buttons
- Arrows where required
- Short explanation

Step-by-Step Explanation

Complex configurations are divided into smaller steps. Flow Designer, for example, is documented as:

Create Flow → Configure Trigger → Get Variables → Create Record → Send Email → Ask for Approval
→ If/Else → Update Record → Test

Replicability

The documentation provides the required information to recreate the solution, including:

- Navigation paths
- Table and field names
- Catalog and variable details
- Flow actions
- Approval conditions
- Testing steps
- Screenshots and expected results

Presentation Structure

The final presentation follows:

Introduction → Business Objective → Functional Scope → Architecture → Configuration →
Automation → Approval → Testing & Security → Demo → Final Outcome

Clarity Checklist

Area	Standard
Headings	Clear and descriptive
Navigation	Exact ServiceNow path
Steps	Sequential and numbered

Area	Standard
Screenshots	Labelled and annotated
Configuration	Required values specified
Verification	Expected result included
Outcome	Result clearly stated
Replicability	Sufficient setup information

5.10 Realism & Quality

The **Automated Network Request Management** solution is designed to represent a realistic enterprise IT service request process. It includes request validation, approvals, automated processing, notifications, status tracking, fulfilment, and closure.

The solution was tested using realistic scenarios and sample data to verify the complete request lifecycle.

Enterprise-Style Request Handling

The request lifecycle follows:

Request Submission → Validation → Record Creation → Approval → Fulfilment → Notification → Closure

This separates the responsibilities of requesters, approvers, fulfilment teams, and administrators.

Role-Based Activities

Requester

- Submits the request.
- Provides required information.
- Tracks request status.
- Receives notifications.

Approver

- Reviews the request.
- Approves or rejects the request.

Network Fulfilment Team

- Processes approved requests.
- Updates fulfilment status.

- Completes the request.

IT Administrator

- Maintains configurations.
- Monitors flows and logs.
- Manages access and troubleshooting.

Data Validation

The solution uses:

- **Mandatory fields** to ensure required information is provided.
- **Conditional fields** such as Existing ID and Please Specify.
- **Approval validation** to control approval-dependent processing.
- **Data mapping validation** to confirm catalog values reach the Network Database.

Realistic Use Cases

Use Case 1 – New Connection

Submit Request → Approval → Approval Decision → Fulfilment → Closure

Use Case 2 – Existing Connection

When Type of Connection = Existing, the Existing ID field is displayed and captured in the request.

Use Case 3 – Other Device

When Types of Devices = Others, the Please Specify field is displayed.

Use Case 4 – Approval Required

Request Submitted → Approval Generated → Approved → Processing

Use Case 5 – Approval Rejected

Request Submitted → Approval Generated → Rejected → Record Updated → Requester Notified

5.10.5 Sample Test Data

Field	Sample Value
Requested For	Test User
Mobile Number	Test Mobile
Type of Connection	Existing
Existing ID	NET-TEST-001
Types of Devices	Others

Field	Sample Value
Please Specify	Network Router
Total Amount	5000
Mode of Payment	UPI
Address	Test Office Location
Urgency	Medium
Business Justification	Network connectivity required

Quality Verification

Quality Area	Verification
Form	Required and conditional fields work correctly
Data	Values are captured and mapped correctly
Approval	Requests follow the configured approval path
Automation	Flow actions execute correctly
Notification	Configured emails are generated
Security	Access controls are reviewed
Status	Request status updates correctly
Auditability	Approval and activity history is available
Closure	Completed requests reach the final state

End-to-End Quality Flow

Test Scenario → Submit Request → Validate Data → Create Record → Verify Approval → Test Decision → Verify Flow → Check Notification → Update Status → Fulfilment → Closure

This validation confirms that the solution functions consistently across the complete request lifecycle.

5.11 Scalability & Future Plan

The **Automated Network Request Management** solution uses standard ServiceNow components and a structured design, making it suitable for future expansion. The existing catalog, approval, workflow, and security structure can be extended as business requirements grow.

Future Enhancement Architecture

The future architecture can follow:

Service Portal → Service Catalog → Network Request → Flow Designer → Approval → Integration Hub → Network Automation → Status Update → Notification & Reporting

Network Automation

Future versions can integrate approved network automation tools such as **Cisco DNA Centre**, **Ansible**, or other organization-approved platforms.

Possible Flow:

Request → Approval → Integration → Network Automation → Result → ServiceNow Status Update

This can reduce manual fulfilment activities where automated provisioning is appropriate.

Integration Hub

Integration Hub can connect ServiceNow with external systems to:

- Send request information.
- Trigger external automation.
- Receive processing results.
- Update ServiceNow records.
- Synchronize status information.

Reporting and Dashboards

Future dashboards can provide visibility into:

- Total requests
- Pending approvals
- Requests by type
- Requests in fulfilment
- Completed and rejected requests
- Average fulfilment time

- Open versus closed requests

These reports can help administrators monitor request volume and service performance.

Expansion to Other IT Services

The same architecture can be extended to other IT services, such as:

- **Device Requests:** Laptops, desktops, monitors
- **Access Requests:** Application and folder access
- **Software Requests:** Software installation and licenses
- **Infrastructure Requests:** Servers, cloud resources, and storage

Reusable Model:

Service Catalog → Request Form → Validation → Approval → Automation → Fulfilment → Notification → Closure

Scalability Considerations

The solution can be expanded by adding:

- Additional catalog items and variables
- New approval rules
- Additional fulfilment groups
- Integration actions
- Automated provisioning
- Dashboards and reports
- Additional security controls

Future Roadmap

Short Term

- Improve catalog usability.
- Add request types.
- Enhance notifications and reports.

Medium Term

- Develop dashboards.
- Add Integration Hub flows.
- Automate selected fulfilment tasks.

Long Term

- Implement network provisioning.

- Integrate network automation platforms.
- Expand to additional IT services.
- Introduce broader enterprise automation.

Future-State Example

User Request → Validation → Approval → Integration Hub → Network Automation → ServiceNow Update → Notification → Closure

This future model can provide greater automation while building on the existing ServiceNow architecture.

Project Conclusion

The **Automated Network Request Management** project demonstrates how ServiceNow can be used to manage network service requests through a structured and automated lifecycle.

The solution captures request information through the **Service Portal and Service Catalog**, using mandatory fields and dynamic form behavior to improve data collection. **Flow Designer** automates record creation, approvals, notifications, and status updates.

The implementation also supports role-based approval processing, allowing requests to be reviewed before fulfilment. Request status, approval records, email notifications, activity history, and system logs provide visibility and traceability throughout the lifecycle.

The solution uses standard ServiceNow capabilities, including:

- Service Portal
- Service Catalog
- Catalog Variables and Variable Sets
- Custom Tables
- UI Policies
- Flow Designer
- Approval Actions
- Email Notifications
- ACLs
- System Logs and Audit History

The project was validated through test scenarios covering request submission, dynamic fields, data mapping, approval processing, notifications, status updates, and closure.

Overall Outcome

The project provides a centralized and structured approach to network request management by combining **request intake, approval, fulfilment, communication, and tracking** within ServiceNow.

Future enhancements can include **Integration Hub, network automation, automated provisioning, dashboards, and additional IT service catalog items**, providing a foundation for further service management automation.

Final Statement

Overall, the **Automated Network Request Management** project demonstrates how ServiceNow can transform a manual network request process into a **structured, traceable, and automated IT service workflow** with scope for future expansion.